

Basics of linux

Some commands explained:

- **ls:** lists all the files inside the directory you are using right now
- **pwd:** stands for print working directory, it prints the paths of the folder you are currently in.
- **man:** is a helpful command and it stand for manual it prints what can be done by a certain command. For instance, man ls → shows what commands or options can be used with ls.
- **Touch:** is used to create a new file. For example, touch text.txt creates a new text file
- **cd:** lets you navigate to a certain directory
- **cd .. :** makes you go back one step in the directory
- **mv:** you can use it to move or rename a file.
- **less:** Opens a file in a scrollable viewer, supports searching, doesn't exit at end only exits on pressing q.
- **more:** the same as "less" command but it exits automatically at the end of the file
- **cat:** stand for concatenate it prints the files for reading, support the reading of multiple files at the same time. It dumps everything on the screen up down.
- **tac:** the opposite of cat it dumps everything on the screen bottom to up, reverses the lines but never the words/sentences
- **rm:** deletes files or directories
- **clear:** just clears the terminal, does not read, write, or modify content.
- **Mkdir:** it creates a new directory/folder

Linux Permissions & Rights:

In this topic we have 2 very important commands

- **sudo chown :** here you write who is the owner and what is the group name. for example, sudo chown Beshoy : BBS text.txt → this here sets the owner to Beshoy and the group of the text.txt into BBS
- **chmod:** in linux we have the permission as 3 different numerical values
read: counts as 4
write: counts as 2
execute: count as 1
no permission: 0

so something like 775 → rwxrwxrw- is a typical number that we will see many time. It gives the owner and the group all privileges and the others only to read and execute

an example command would be `chmod 775 text.txt`

Answers to exercise questions:

Q: What types of permissions are distinguished between user and group rights in Linux/Unix? What letters are used for the permissions? What bit is used for the permissions?

A: we have 3 types of ownership Owner, Group, Others. We have 3 letters each represents a type of privilege which is the permission r = read, w = write, x = execute. Each group uses 3 bits.

Each of them has a numerical value. r= 4 , w=2, x= 1

Q: What is the command to make a file readable and executable only for the user? Provide this command both with arguments consisting of numbers and consisting of characters.

A: `chmod 500 filename` → numerical `chmod u=r-w, go= filename`

Q: some commands for the exercise:

cd / : Navigate to the home directory

mkdir Practice1: Creates a directory in the directory where you are standing right now

touch Practice1/wurzel.txt: Creates a text file called wurzel in the Practice1 directory

ls -l / > Practice1/wurzel.txt: the `ls -l` lists all the information the `/` makes this listed information about the home directory. The symbol `>` redirects this data into a file. Then we list the file we want the data to be directed to

nano Practice1/wurzel.txt: nano is a read write type of command

cat Practice1/wurzel.txt: as discussed before the cat lists the data (dumps) it to be read

ll Practice1/wurzel.txt: ll just lists all the data about the selected file including the size and the permission.